Scale-Consistent Multispectral Reconstruction From UAV and Satellite Observations for Coastal Bathymetric Inversion

Jianchun Chen, Ya Ping Wang, Yan Gu, Shibing Zhu, Chao Gao, and Renzhi Li

***Abstract*—High-resolution quantitative remote sensing is constrained by the spatial–spectral trade-off among current observation platforms: uncrewed aerial vehicle (UAV) imagery provides fine spatial detail but limited spectral dimensionality, whereas satellite multispectral data provide richer spectral information at coarser spatial scales. This study proposes data-driven spectral-structural anchoring fusion (DSSAF), a scale-consistent framework that reconstructs UAV-scale multispectral imagery by integrating fine-resolution structural information from UAV RGB imagery with multispectral constraints from PlanetScope observations. The framework combines land-cover-conditioned spectral mapping, data-driven structural reference selection, adaptive detail injection, and scale-consistency correction. It was evaluated over a heterogeneous intertidal sandbar system and further tested through shallow-water bathymetric inversion. The reconstructed imagery preserved UAV-scale textures, gradients, and edges while remaining radiometrically consistent with the original PlanetScope data after reaggregation to the native satellite scale. In the downstream bathymetry task, DSSAF-fused imagery achieved the best performance among the tested inputs, with $R^2 = 0.924$ and RMSE = 0.079 m. These results demonstrate that scale-consistent UAV-satellite fusion can improve both image-level spatial–spectral representation and quantitative geophysical retrieval, providing a transferable approach for high-resolution environmental mapping in heterogeneous coastal landscapes.**

***Index Terms*—Bathymetric inversion, cross-scale fusion, multispectral reconstruction, PlanetScope, scale consistency, uncrewed aerial vehicle (UAV).**

I. Introduction

A FUNDAMENTAL limitation in quantitative remote sensing is that no single optical sensor generally provides both fine spatial detail and rich spectral information. Spatial resolution controls the representation of object boundaries, textures, and microtopographic features, whereas spectral resolution supports the retrieval of surface composition, moisture, chlorophyll status, and environmental gradients [1], [2], [3], [4], [5]. Because sensor design and platform constraints impose a persistent trade-off between spatial and spectral resolution, reconstructing multispectral information at a fine and consistent spatial scale while preserving authentic spatial structures remains a central problem in remote sensing data fusion [6], [7], [8], [9], [10].

Satellite observation systems provide a stable foundation for long-term monitoring at regional to global scales [11], [12]. Data streams from Landsat, Sentinel-2, and PlanetScope have mature processing ecosystems for radiometric and geometric correction, time-series analysis, and large-area mapping, enabling consistent regional comparison and temporal monitoring [13]. However, when target features contain subtle transition zones, complex boundaries, or high-frequency local processes, pixel sizes from meters to tens of meters inevitably introduce mixed-pixel effects [14], [15]. These effects weaken texture representation and microstructural expression, leading to blurred boundaries, reduced feature separability, and lower performance in fine-scale classification, feature extraction, and parameter inversion [16].

Uncrewed aerial vehicle (UAV) platforms complement satellite observations through rapid deployment, flexible acquisition timing, and centimeter-level spatial detail [17], [18]. In practice, however, many UAV surveys still rely on RGB cameras, which limits their use in quantitative applications that require near-infrared (NIR) or red-edge information [1]. Examples include multispectral-index-based land-cover discrimination, surface-moisture characterization, and optical inversion in shallow or turbid waters [19], [20]. Therefore, supplementing UAV observations with physically meaningful multispectral information without sacrificing fine spatial detail is essential for making UAV remote sensing more interpretable, transferable, and quantitatively useful [21].

Spatio-spectral fusion addresses this need by reconstructing high-resolution multispectral products that combine fine spatial details from UAV imagery with spectral information from satellite observations [22], [23]. Existing approaches include component substitution and multiresolution analysis for detail injection, Bayesian methods with explicit degradation models and regularization, sparse-representation and matrix-decomposition methods based on low-dimensional priors, and deep learning models that learn end-to-end super-resolution mappings [24], [25], [26].

Despite these advances, direct application to UAV RGB and PlanetScope multispectral fusion remains challenging [27], [28]. First, UAV RGB imagery and PlanetScope data differ substantially in spectral response functions, band configurations, spatial resolution, and point spread functions. Second, UAV imagery is highly sensitive to illumination, bidirectional reflectance distribution function (BRDF) effects, shadows, and

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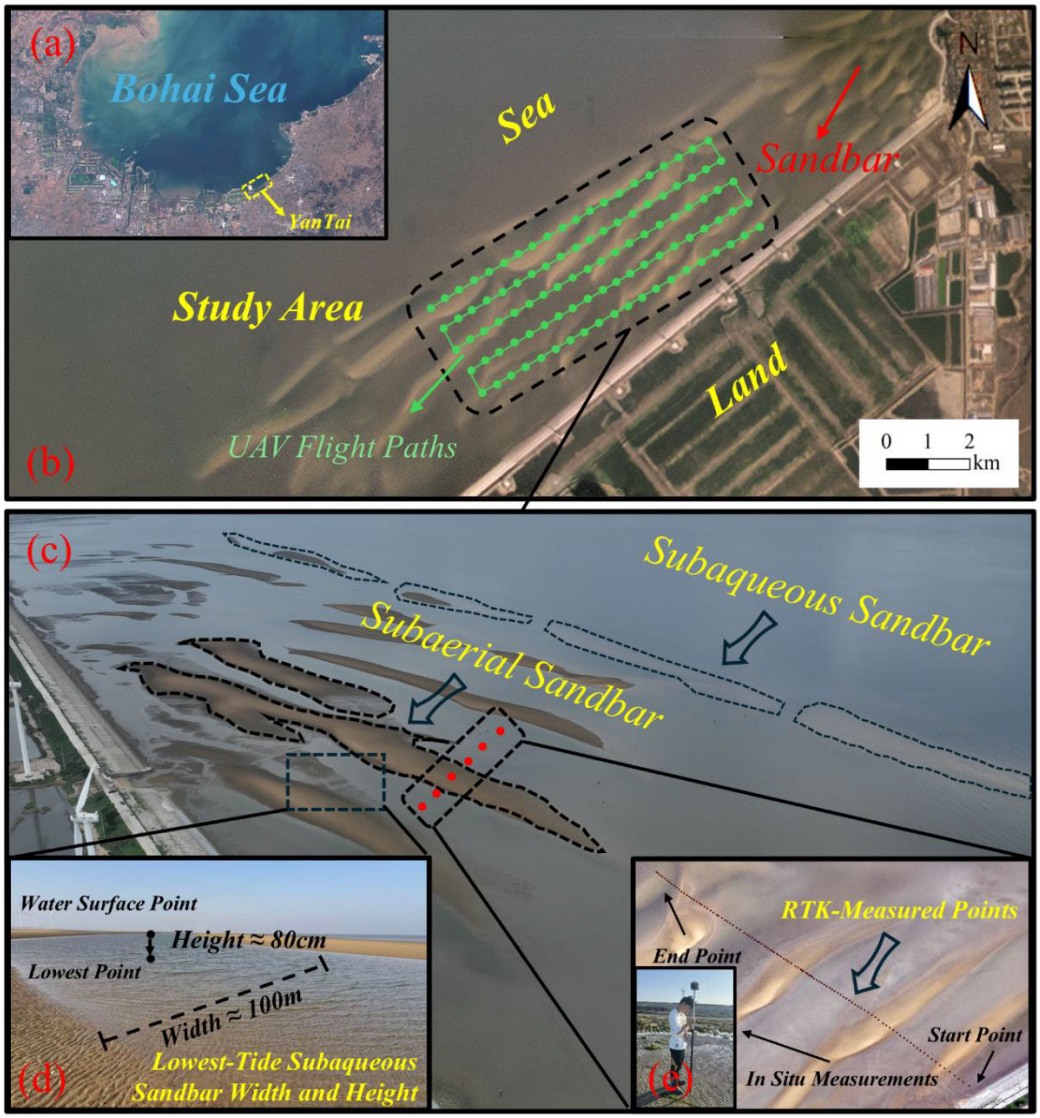


Fig. 1. Location of the study area, sandbar distribution, and field survey design. (a) Location of the study area in Bohai Bay (Yantai, Shandong Province); (b) remote sensing image showing the spatial extent and distribution of sandbars in the study area; (c) example of UAV orthomosaic with exposed sandbars, submerged sandbars, and RTK measurement points annotated; (d) schematic illustration of submerged sandbar width and relative water level during low tide; (e) layout of RTK survey points and photographs from the field measurements.

viewing geometry, so a fixed degradation model is unlikely to remain valid across heterogeneous coastal surfaces [29], [30]. Third, cross-scale registration errors, class-dependent nonlinearities, and mixed pixels can amplify artifacts or cause spectral drift during detail injection [31], [32], [33]. A practical framework is therefore needed to decouple spectral mapping from structural enhancement while explicitly constraining the fused result at the native satellite scale [34].

To address these challenges, this study proposes data-driven spectral–structural anchoring fusion (DSSAF), a scale-consistent cross-sensor framework for high-resolution multispectral reconstruction from UAV and PlanetScope observations. The main contributions are fourfold. First, a land-cover-conditioned spectral mapping strategy is introduced to reduce class-dependent bias and land-cover mixing effects. Second, a data-driven structural reference selection mechanism is developed to identify stable UAV-scale spatial components for detail transfer. Third, adaptive structural detail injection is coupled with scale-consistency correction to enhance fine spatial information while suppressing low-frequency spectral drift. Fourth, reconstruction performance is evaluated not only through image-level spectral and spatial metrics but also through a downstream shallow-water bathymetric inversion task. Beyond the UAV-PlanetScope configuration examined here, DSSAF is designed as a transferable framework for integrating structural and spectral information across observation scales.

II. Study Area and Datasets

A. Study Area

The study area is located in the nearshore waters of Yantai, Shandong, China, and represents a typical intertidal sandbar system (Fig. 1). The region lies along the southwestern Bohai Sea and is jointly influenced by Yellow River runoff, tidal forcing, and wave action. Active nearshore sediment transport has formed a well-developed sandbar-tidal creek system. Sandbars are distributed along the coastline in belt-like or elongated patterns and exhibit pronounced surface elevation variability. Their upper portions remain exposed for most of

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the year or are only briefly inundated during spring tides, whereas their lower portions are frequently submerged during normal tidal cycles. This configuration creates strong gradients in inundation, sediment properties, and surface morphology, making the area suitable for evaluating the integration of fine-scale structural information and multispectral observations.

The regional tidal regime is characterized by irregular semidiurnal tides, with an average tidal range of approximately 0.2–2 m and a maximum spring tidal range of about 5 m. Periodic inundation and exposure, combined with wave- and current-driven sediment transport, produce complex spatial structures and strong spectral variability across multiple spatial scales. Surface sediments are dominated by well-sorted medium to fine sand, locally mixed with silty material. The open coastline is also strongly affected by prevailing winds; during mid- to high-water stages, wind-driven waves can reshape both the sandbars and the landward tidal creeks. These coupled hydrodynamic and sedimentary conditions provide a representative coastal setting for testing cross-scale multispectral reconstruction under heterogeneous conditions.

B. UAV RGB Dataset

UAV imagery was acquired using a DJI Phantom 4 series quadcopter equipped with a 1-in CMOS sensor that records high-resolution imagery in the red, green, and blue visible bands. The aerial survey was conducted on August 1, 2025, during a low-tide stage at a flight altitude of approximately 160 m, corresponding to a ground sampling distance of about 0.05 m.

To balance spatial coverage and image overlap, parallel flight paths were planned with forward and side overlaps of no less than 80% and 70%, respectively. Meteorological conditions during the survey were favorable, with low cloud cover and high visibility, ensuring stable surface reflectance acquisition.

C. PlanetScope Multispectral Dataset

Multispectral satellite data were obtained from the PlanetScope constellation as Level 3B surface reflectance products. Scenes covering the study area and acquired close in time to the UAV survey were selected. The PlanetScope data have a spatial resolution of approximately 3 m and include eight visible to NIR bands: Coastal Blue, Blue, Green I, Green, Yellow, Red, Red-edge, and NIR. The products were atmospherically corrected and included surface reflectance values together with quality masks for clouds, cloud shadows, and invalid pixels.

D. RTK Dataset

High-precision *in situ* measurements were collected using RTK-GNSS to support sandbar topographic reconstruction and water-depth inversion. Field surveys were conducted on August 1, 2025 under tidal and temporal conditions as consistent as possible with the remote-sensing acquisitions. The measurements included elevation points along representative sandbar profiles and water-depth observations in the shallow-water zones adjacent to the sandbars. Data were collected using a South Surveying RTK receiver, with horizontal and vertical accuracies better than ± 0.02 and ± 0.03 m, respectively.

III. Methodology

A. Data Preprocessing

The proposed spatio-spectral fusion framework uses near-synchronous UAV RGB imagery and PlanetScope multispectral data acquired under clear-sky conditions, weak wave activity, and comparable tidal stages. To support pixel-level fusion and subsequent water-depth inversion, all datasets were processed to improve geometric and radiometric consistency. The preprocessing workflow comprised three steps.

1) *UAV Image Preprocessing:* Raw UAV imagery was processed using a structure-from-motion (SfM) and multiview stereo (MVS) workflow implemented in Pix4Dmapper. During bundle adjustment, exterior orientation parameters were jointly optimized, and lens distortion was corrected using internal software calibration or external calibration information. Dense image matching was then performed to generate point clouds, which were rasterized to produce a high-resolution digital elevation model (DEM) and orthorectified imagery.
2) *PlanetScope Preprocessing:* Level 3B surface reflectance scenes covering the study area and acquired within one day of the UAV survey were selected. Scenes with cloud cover exceeding 10% or affected by obvious haze were excluded. The surface reflectance products were further reprojected to the coordinate system of the UAV orthomosaic.
3) *Image Co-Registration and Feature Normalization:* To establish spatially consistent correspondence between the PlanetScope multispectral imagery and the UAV orthomosaic, both 3 m and 5 cm grids were constructed. At the 3 m scale, UAV RGB imagery was aggregated to the PlanetScope pixel scale using block averaging, so that aggregated UAV features and PlanetScope pixels shared the same spatial support. These data were used for cross-sensor relationship learning, spatial structural reference selection, and training of band-specific spatial proxy models.

At the 5 cm scale, each PlanetScope band was resampled to the UAV grid using bilinear interpolation, providing low-frequency multispectral reflectance features for each UAV pixel. After co-registration and valid-pixel screening, the first and 99th percentiles of each UAV and PlanetScope band were calculated within the valid study area, and linear scaling was applied to map each feature to the [0, 1] range. This normalization reduced the influence of intersensor radiometric range differences, outliers, and extremely bright or dark pixels during model training.

B. Land-Cover Stratification

A supervised machine-learning land-cover stratification module was introduced before spatial–spectral fusion to reduce the influence of class-dependent spectral responses, spatial textures, and cross-sensor mapping relationships [35]. According to the dominant surface characteristics of the intertidal study area, four land-cover classes were defined: water/shallow water, wet sand, dry sand/exposed sandbar, and vegetation/background. The resulting subregions exhibit relatively homogeneous spectral and spatial characteristics and therefore provide class-specific constraints for subsequent spatial-proxy modeling and adaptive structural-detail injection.

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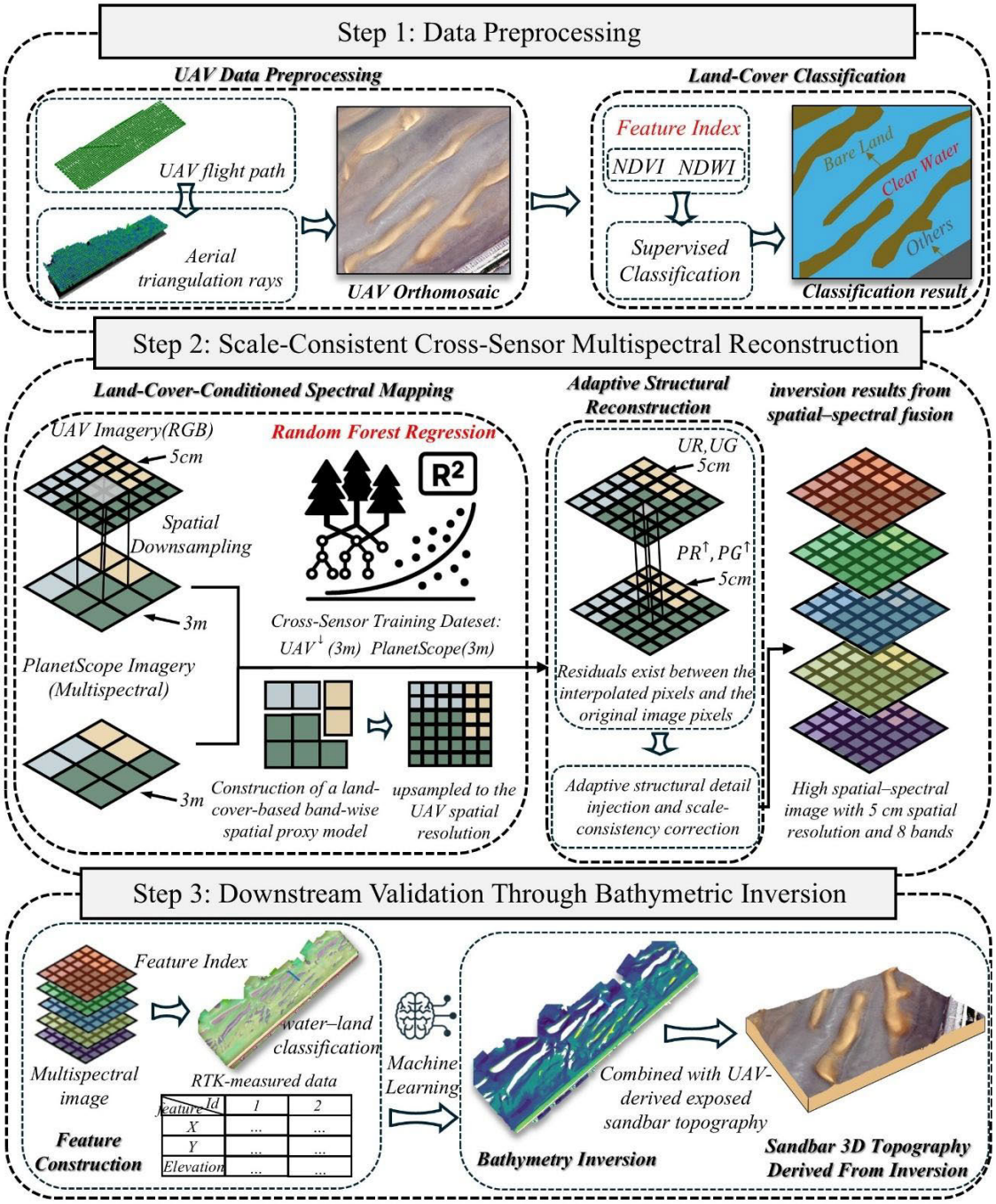


Fig. 2. Technical flowchart.

Representative training samples were selected from the co-registered UAV–PlanetScope imagery through visual interpretation and interactive sampling, and each sample was assigned to one of the four land-cover classes. The classification feature set consisted of the PlanetScope Blue, Green, Red, and NIR bands together with three derived features: the normalized difference water index (NDWI), normalized difference vegetation index (NDVI), and visible-band brightness. NDWI and NDVI were calculated from the Green–NIR and NIR–Red band combinations, respectively, whereas brightness was defined as the mean of the Blue, Green, and Red bands. These spectral bands and derived indices jointly characterize water occurrence, vegetation response, surface moisture, and overall surface brightness, thereby improving the separability of the dominant intertidal surface types. A Random Forest classifier was then trained using these features, as expressed in the following equation:

$$L_i = C_{RF}(S_i) \tag{1}$$

where L_i denotes the land-cover class of pixel i, C_{RF} represents the random forest classification model, and S_i is the classification feature set. The Random Forest classifier consisted of 300

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TABLE I
SENSITIVITY OF STRUCTURAL REFERENCE SELECTION TO DIFFERENT WEIGHTING SCHEME

Weighting Scheme	Radiometric Consistency	High-frequency Consistency	Gradient Consistency	Selected structural reference	Stability score
W1	0.50	0.30	0.20	Red	0.529
W2	0.45	0.30	0.25	Red	0.571
W3	**0.40**	**0.35**	**0.25**	**Red**	**0.613**
W4	0.35	0.40	0.25	Red	0.582
W5	0.35	0.35	0.30	Red	0.558
W6	0.33	0.33	0.34	Red	0.547

decision trees, with no explicit restriction on the maximum tree depth. The minimum number of samples required to split an internal node was set to 2, whereas the minimum number of samples required at each leaf node was set to 1. At each node, the number of candidate features considered for the optimal split was set to the square root of the total number of input features. Bootstrap sampling was enabled, and a fixed random seed of 42 was used to ensure reproducibility. Parallel tree construction was performed using all available CPU cores.

After training, the classifier was applied to the entire study area to generate a land-cover stratification map. This classification result was not used to modify the original radiometric values of the imagery, nor was it intended as the final mapping product of this study; instead, it served as an auxiliary constraint in the spatial–spectral fusion process. On the one hand, band-specific spatial proxy models were trained separately within different land-cover classes to reduce the mapping bias introduced by a single global model over heterogeneous surfaces. On the other hand, the land-cover stratification results were used to regulate the intensity of spatial detail injection in different classes. As a result, regions with clear textures and stable structures were allowed to incorporate stronger UAV-scale details, whereas regions with complex spectral variation or unstable structure were treated with a more conservative detail-enhancement strategy, thereby reducing pseudo-texture amplification and class confusion.

C. Selection of Spatial Structural Reference Components Under Cross-Sensor Consistency Constraints

After land-cover stratification, stable high-resolution spatial structural reference components were identified to guide detail enhancement of the multispectral target bands. Because individual UAV visible bands and their composite features respond differently to surface texture, edges, and brightness variations, no single UAV band can be assumed a priori to be the optimal reference. Therefore, both original UAV bands and derived composite features were treated as candidate structural references and were evaluated under cross-sensor consistency constraints.

Each candidate structural reference component was first aggregated to the native multispectral scale so that it shared the same spatial support as the corresponding PlanetScope component. Candidate stability was evaluated using three criteria: radiometric consistency, high-frequency structural consistency, and gradient consistency. Radiometric consistency measured overall numerical agreement, high-frequency structural consistency measured similarity in local texture and fine-scale variation, and gradient consistency measured correspondence along boundaries and spatial transitions. These three criteria were selected because they describe complementary aspects of cross-sensor structural similarity. Radiometric consistency reflects the overall correspondence between sensors, high-frequency consistency represents local texture similarity, and gradient consistency characterizes edges and spatial transitions. Their combination therefore provides a more complete assessment of whether a candidate component can serve as a stable structural reference than any single metric alone. These metrics were integrated into a stability score, as shown in the following equation:

$$S_j = \omega_1 C_j^{rad} + \omega_2 C_j^{HF} + \omega_3 C_j^{grad} \tag{2}$$

where S_j denotes the overall score of the jth candidate spatial reference component; C_j^{rad}, C_j^{HF}, C_j^{grad} represent radiometric consistency, high-frequency structural consistency, and gradient consistency, respectively; and ω_1, ω_2, and ω_3 are the corresponding weights. In this study, the weights assigned to radiometric consistency, high-frequency structural consistency, and gradient consistency were set to 0.40, 0.35, and 0.25, respectively. To further assess the robustness of the weighting scheme, a sensitivity analysis was conducted using several alternative combinations of the three coefficients while maintaining their sum equal to one (Table I). The UAV Red component remained the highest-ranked structural reference under all tested weighting schemes, indicating that the reference selection was relatively stable to moderate variations in the weighting coefficients and was not dependent on a specific parameter combination. The candidate component with the highest overall score was ultimately selected as the spatial structural reference for subsequent detail injection. The core purpose of this step is to establish spatial detail transfer on the basis of a relatively stable structural correspondence between the high-spatial-resolution imagery and the multispectral imagery, thereby reducing the uncertainty associated with manually specifying a single reference band. This structural correspondence provides a stable basis for transferring fine-scale spatial information across observation scales while maintaining consistency with the multispectral observations.

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D. Construction of Band-Specific Spatial Proxy Models

After selecting the spatial structural reference component, a separate spatial proxy model was constructed for each target multispectral band. The purpose of this step was not to generate the final fused image directly, but to predict a high-resolution proxy for each multispectral band that preserves band-specific spatial variation. In contrast to methods that inject the same high-resolution texture into all bands, band-specific modeling helps maintain interband differences and reduces the risk of cross-band texture homogenization and pseudo-detail amplification.

Specifically, at the native spatial scale of the multispectral imagery, aggregated high-spatial-resolution image features, spatial structural reference components, brightness features, local texture features, and gradient features were used as inputs, while the corresponding multispectral target band was used as the output to train separate random forest regression models. For pixel i, the input feature vector is defined as shown in the following equation:

$$x_i = [U_i, A_i, Br_i, T_i, G_i] \tag{3}$$

where U_i denotes the original band features of the high-spatial-resolution imagery at pixel i, A_i represents the spatial structural reference component selected in the previous step, Br_i denotes the brightness feature, T_i represents the local texture feature, and G_i denotes the gradient feature. For the kth multispectral target band, the supervisory label is the original observation of that band at the same location, namely the normalized reflectance of the kth multispectral band at pixel i, denoted as $P_{k,i}$. Therefore, the spatial proxy model for the kth band can be defined as shown in the following equation:

$$P_{k,i} = f_k(x_i) + \varepsilon_i \tag{4}$$

where $f_k(\cdot)$ denotes the random forest regression model corresponding to the kth target band, and ε_i represents the residual term. The same Random Forest configuration was used consistently for all band-specific spatial-proxy models. Each model consisted of 300 regression trees with unrestricted maximum tree depth. The minimum number of samples required to split an internal node was set to 2, and the minimum number of samples at each leaf node was set to 1. All predictor features were considered when searching for the optimal split, and bootstrap sampling was enabled. A fixed random seed of 42 was adopted to ensure reproducibility, and all available CPU cores were used for parallel computation. To balance computational efficiency and sample representativeness, the number of training pixels was limited to a maximum of 300 000 valid samples per land-cover class. All machine-learning procedures were implemented in Python 3.11 using scikit-learn 1.5.2.

After model training, the same feature-construction procedure was applied to the high-spatial-resolution image grid to generate the spatial proxy image of the kth target band at the target fusion scale, as shown in the following equation:

$$Q_k^{HR} = f_k\left(x^{HR}\right) \tag{5}$$

where Q_k^{HR} denotes the high-resolution spatial proxy result for the kth multispectral band. To reduce mapping bias among different land-cover types, the previously derived machine learning–based land-cover classification results were incorporated. Band-specific random forest models were trained or applied separately within different land-cover classes, enabling the models to adapt to class-dependent differences in spectral–spatial relationships. It should be noted that Q_k^{HR} is not directly used as the final fused band. Instead, it is used to extract the potential high-frequency spatial details of the target band and to provide a band-specific spatial basis for subsequent adaptive structural detail injection. Therefore, the proxy model serves as an intermediate representation that links multispectral observations and fine-scale spatial structures during the reconstruction process.

E. Adaptive Structural Detail Injection and Scale-Consistency Correction

After obtaining the high-resolution spatial proxy image Q_k^{HR} for the kth target band, band-specific high-frequency spatial details were further extracted and adaptively fused with the detail information from the spatial structural reference component. Specifically, the spatial proxy image was first aggregated back to the native spatial scale of the multispectral imagery and then upsampled again to the target fusion scale to obtain its low-frequency component. The proxy detail of the target band was then derived by subtracting this low-frequency component from the original spatial proxy image, as shown in the following equation:

$$D_k^{proxy} = Q_k^{HR} - Up\left(Down\left(Q_k^{HR}\right)\right) \tag{6}$$

where $Down(\cdot)$ denotes the block-averaging operation that aggregates data from the target fusion scale to the native multispectral scale, and $Up(\cdot)$ denotes the upsampling operation from the multispectral scale to the target fusion scale. Meanwhile, high-spatial-resolution details were extracted from the spatial structural reference component A^{HR}, as shown in the following equation:

$$D^{ref} = A^{HR} - Up\left(Down\left(A^{HR}\right)\right) \tag{7}$$

where DD^{ref} provides texture, edge, and local structural information from the true high-spatial-resolution imagery. To avoid applying the same detail-injection intensity to all multispectral bands, an adaptive injection weight was calculated according to the spectral correlation, high-frequency structural correlation, and gradient correlation between the target band and the spatial reference component, as shown in the following equation:

$$\omega_k = Norm\left(0.10\rho_k^{spec} + 0.55\rho_k^{HF} + 0.35\rho_k^{grad}\right) \tag{8}$$

where ρ_k^{spec}, ρ_k^{HF}, and ρ_k^{grad} denote the spectral correlation, high-frequency structural correlation, and gradient correlation between the kth target band and the spatial reference component, respectively. $Norm(\cdot)$ represents a normalization function that maps the integrated similarity score to a predefined weight interval. Because the validity of detail injection primarily depends on the transferability of spatial structures, higher weights were assigned to high-frequency structural correlation and gradient correlation. The mixed detail for the target band was then obtained as follows:

$$D_k^{mix} = (1 - \lambda_k) D_k^{proxy} + \lambda_k D^{ref} \tag{9}$$

where λ_k is jointly determined by ω_k and the reference-detail proportion. To further adapt to spatial–structural

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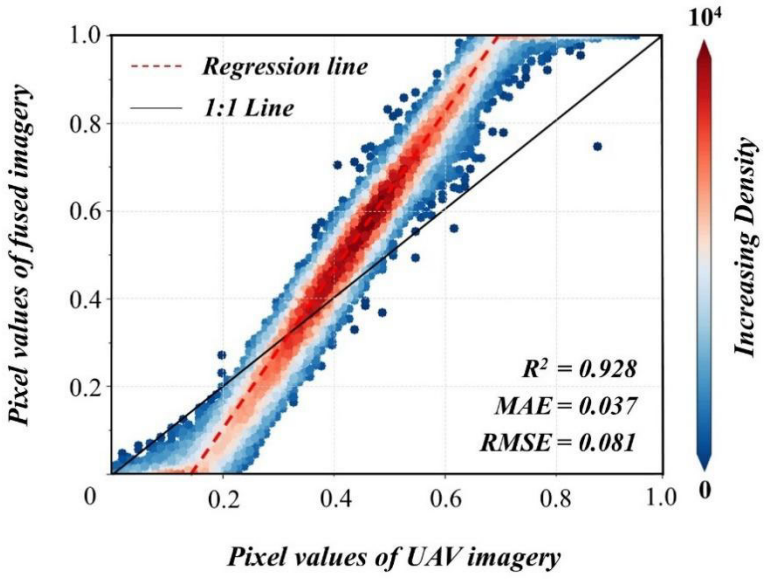


Fig. 3. Density scatter comparison between the fused band and the UAV reference values.

differences among land-cover types and target bands, class-specific gains and band-specific gains were introduced based on the land-cover classification results and band similarity. This allows regions with clear textures and stable structures to incorporate stronger spatial details, whereas areas with complex spectral variations or unstable structures are treated with a more conservative detail-enhancement strategy.

During detail injection, the kth multispectral band was first upsampled to the target fusion scale to form a low-frequency spectral basis P_k^{up}, as shown in the following equation:

$$P_k^{up} = Up\left(P_k\right). \tag{10}$$

The mixed detail was then injected into this low-frequency basis to generate the initial fused result F_k^{raw}, as shown in the following equation:

$$F_k^{raw} = P_k^{up} + D_k^{mix}. \tag{11}$$

To prevent detail injection from shifting the low-frequency spectral distribution of the multispectral band, a scale-consistency correction was further introduced. Specifically, the initial fused result was aggregated back to the native multispectral scale, and the difference between this aggregated result and the original multispectral band was calculated. This difference was then upsampled back to the target fusion scale for compensation, as shown in the following equation:

$$F_k^{HR} = F_k^{raw} + Up\left(P_k - Down\left(F_k^{raw}\right)\right). \tag{12}$$

Through this correction, the final fused result remains consistent with the original multispectral observation after being aggregated back to the native multispectral scale, as shown in the following equation:

$$Down\left(F_k^{HR}\right) \approx P_k. \tag{13}$$

Therefore, this step plays two key roles. First, D_k^{proxy} and D^{ref} introduce target-band-related high-resolution texture and edge structures, enabling the reconstruction of fine-scale spatial variability. Second, the scale-consistency correction preserves the low-frequency spectral constraint of the original multispectral imagery, ensuring that the reconstructed product remains consistent with the original multispectral observations when aggregated back to the native satellite scale.

F. Fusion-Driven Shallow-Water Bathymetry

To evaluate the practical information value of different image products for downstream quantitative retrieval, shallow-water bathymetric inversion was used as an application-oriented validation task. Three input scenarios were compared: native PlanetScope imagery, UAV RGB imagery, and DSSAF-fused imagery. The purpose of this comparison was not to artificially equalize the spectral dimensionality of the three products, but to evaluate their respective spatial–spectral information content under otherwise consistent inversion conditions. Therefore, the original spectral bands available from each image product were directly used as predictors, without introducing additional spectral indices or band-ratio features. Differences in spatial resolution and spectral dimensionality were retained as intrinsic characteristics of the respective image products

$$d_i = f\left(\mathbf{z}_i\right) + \varepsilon_i \tag{14}$$

where d_i denotes the measured water depth of the ith sample point, $\mathbf{z}_i$ represents the corresponding feature vector extracted from the fused imagery, $f(\cdot)$ is the random forest regression model, and ε_i is the residual term. To ensure comparability among the three input scenarios, the same RTK reference samples, the same training–testing partition, the same Random Forest configuration, and the same evaluation procedure were used throughout the comparative experiment. The RTK samples were partitioned once at a ratio of 7:3 for model training and testing, respectively, and this identical partition was then applied to the PlanetScope-only, UAV RGB-only, and

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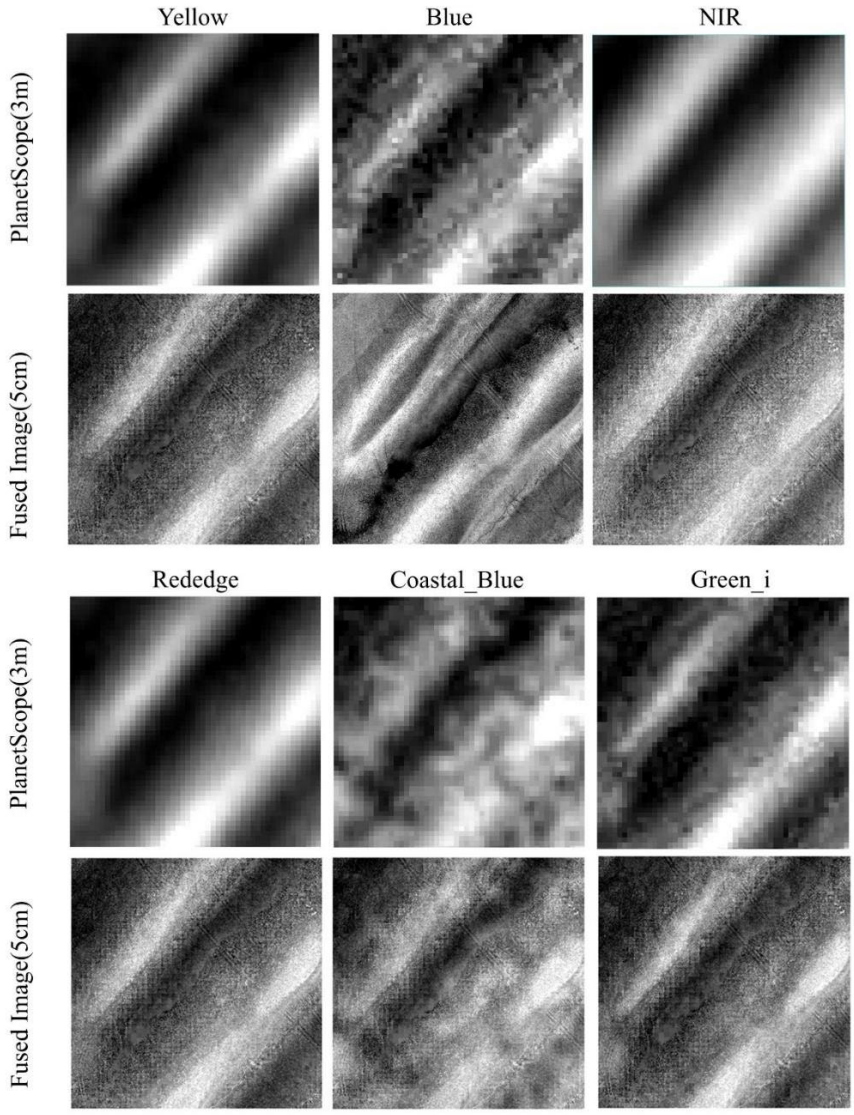


Fig. 4. Spatial details and cross-band differences of multispectral spatiospectral fusion. Local texture comparison between the original PlanetScope imagery (3 m) and the fused imagery (5 cm) across multiple spectral bands.

DSSAF-fused experiments. After model training, the corresponding model was applied to the valid shallow-water area to generate a continuous water-depth distribution. Exposed sandbars, nonwater areas, and regions where water-depth inversion was not performed were excluded from model training and accuracy assessment.

The comparative bathymetric experiment was designed to evaluate the practical information value of different image products rather than to equalize their spectral dimensionality. PlanetScope, UAV RGB, and DSSAF-fused imagery were therefore evaluated using their respective available spectral information. For all three experiments, identical RTK reference samples, the same training–testing partition, the same Random Forest configuration, and the same evaluation procedures were used. The training–testing partition was generated once and consistently applied to all three datasets, so that the principal experimental variable was the remote-sensing image product used for bathymetric inversion.

IV. Results

A. Reconstruction Fidelity and Texture Recovery

During structural reference selection, the UAV visible bands and their composite features showed clear differences in cross-sensor stability. After jointly considering radiometric consistency, high-frequency structural consistency, and gradient consistency, the UAV Red component achieved the highest overall stability score (0.613) and was selected as the spatial structural reference for subsequent detail injection. This result indicates that the Red component provides the most stable structural correspondence with the PlanetScope multispectral observations in the study area. The selection of the Red band also has a plausible physical interpretation for the investigated coastal environment. Compared with the Blue and Green bands, which are more strongly affected by water-column scattering and variations in shallow-water optical conditions, the Red band generally provides clearer contrast among water, wet sand, exposed sand, and other surface types because of the strong absorption of red wavelengths by water. Consequently,

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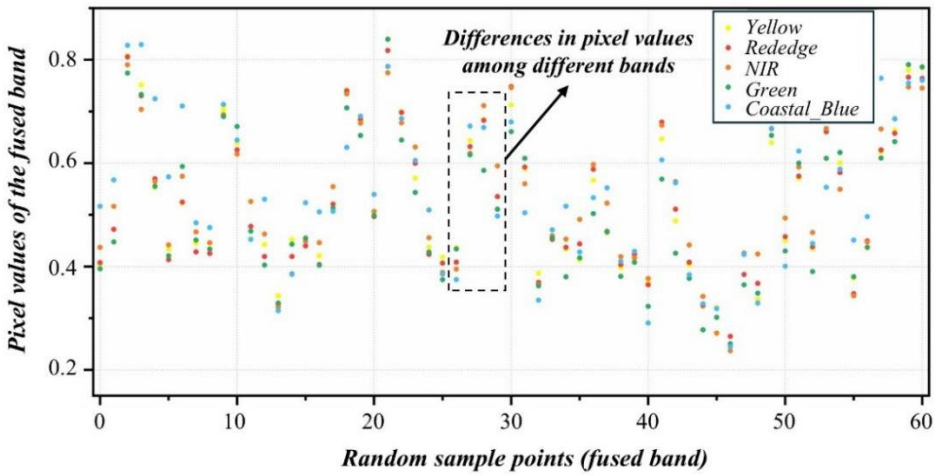


Fig. 5. Distribution of pixel values across spectral bands at randomly sampled locations in the fused imagery, illustrating relative gradients and spectral shape differences among bands.

TABLE II
SCALE-LEVEL SPECTRAL CONSISTENCY ASSESSMENT OF THE FUSED IMAGERY

Band	Spectral Index				
	CC	R^2	RMSE	MAE	SSIM
Coastal Blue	0.999947124	0.999877512	0.002544371	0.000448265	0.999832689
Blue	0.999908929	0.999783874	0.003378935	0.00069303	0.999487714
Green I	0.999785106	0.999415517	0.00594636	0.001943879	0.996804711
Green	0.999709684	0.999158144	0.007354455	0.002798392	0.993696704
Yellow	0.999743349	0.999163866	0.007530011	0.003178779	0.992939638
Red	0.99974316	0.99915874	0.007614459	0.003221532	0.993095524
Red-edge	0.999821795	0.999468267	0.006181312	0.002297645	0.995984236
NIR	0.999866893	0.999635875	0.005246625	0.001637516	0.997275028

boundaries and local spatial transitions associated with sandbars and water–land interfaces are more clearly expressed in the Red band. This spectral behavior is consistent with its higher structural-consistency score in the present study.

To evaluate the reliability of the fused 5 cm product in an observable band, representative areas were randomly selected, and pixel values from the fused imagery were compared with those from the UAV-observed imagery. The scatter points were mainly distributed around the regression line and closely followed the 1:1 line (Fig. 3). Quantitatively, the fused result achieved $R^2 = 0.928$, MAE = 0.037, and RMSE = 0.081 relative to the UAV-observed imagery. These results indicate that the fused imagery effectively preserves UAV-scale spatial–spectral variations in the observable band. Although slight deviations occur in some high-value regions, the overall error remains low, demonstrating that the proposed method can recover fine-scale structural information while maintaining multispectral constraints. Importantly, this comparison provides a direct fine-scale observational assessment that is not determined by the PlanetScope-scale consistency constraint imposed during the final correction step. Because UAV RGB imagery also contributes high-resolution structural information to DSSAF, this comparison is interpreted as complementary fine-scale observational evidence rather than as a fully independent external validation of the entire reconstruction framework.

For PlanetScope bands without corresponding UAV-scale observations, including Coastal Blue, Green I, Yellow, Red-edge, and NIR, direct validation of radiometric accuracy at the 5 cm scale is not possible because independent fine-resolution reference observations are unavailable. Accordingly, these reconstructed bands are interpreted as scale-constrained, structurally informed estimates rather than directly observed centimeter-scale reflectance measurements. The comparisons are therefore used to examine whether the reconstructed bands retain band-dependent radiometric contrast and plausible fine-scale spatial structures, rather than to establish their absolute 5 cm radiometric accuracy (Fig. 4).

The interband comparison further shows that distinct relative gradients and spectral contrast relationships are retained among the reconstructed bands (Fig. 5). This indicates that DSSAF preserves band-specific differences while introducing fine-scale structural information, rather than collapsing the multispectral data into a common spatial texture. The radiometric behavior of these non-UAV-observable bands is evaluated at the native PlanetScope scale through the scale-consistency assessment, whereas their centimeter-scale values should be regarded as model-based estimates.

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TABLE III
SPATIAL STRUCTURAL CONSISTENCY ASSESSMENT OF THE FUSED IMAGERY

Band	Spatial Index			
	High-frequency Correlation	Gradient Correlation	SSIM	Edge F1
Coastal Blue	0.394089157	0.285113244	0.411778666	0.39164162
Blue	0.623157578	0.469004747	0.533958355	0.504474347
Green I	0.781318582	0.610347415	0.648433302	0.605286745
Green	0.811403981	0.619442005	0.667470062	0.615473653
Yellow	0.813657783	0.622932339	0.65314295	0.615152826
Red	0.81238858	0.622498127	0.648456856	0.614853329
Red-edge	0.811742914	0.630359327	0.648462171	0.618819669
NIR	0.754424535	0.571452241	0.579131361	0.572762847

TABLE IV
SPATIAL GAIN OF THE FUSED IMAGERY RELATIVE TO DIRECT UPSAMPLING

Metric	Directly upsampled PlanetScope baseline	Fused result	Spatial gain
High-frequency Correlation	0	0.8604	0.8604
Gradient Correlation	0.011	0.6128	0.6018
Edge F1	0.2849	0.6071	0.3222

To assess whether the reconstructed imagery satisfies the imposed native-scale spectral constraint, the 5 cm fused bands were reaggregated to the native PlanetScope scale and compared with the corresponding original PlanetScope bands (Table II). Across all bands, the mean spectral correlation coefficient and R^2 reached 0.9998 and 0.9995, respectively, while the mean RMSE and MAE were 0.0057 and 0.0020. Because the scale-consistency correction explicitly constrains the aggregated fused imagery toward the original PlanetScope observations, these results are interpreted primarily as evidence that the imposed native-scale spectral constraint is effectively satisfied. Accordingly, the high native-scale agreement is treated as evidence of constraint compliance and limited low-frequency spectral drift, rather than as independent validation of the reconstructed radiometric values at the 5 cm scale. Fine-scale reconstruction quality is assessed separately through comparison with the corresponding UAV-observed band, while the RTK-based shallow-water bathymetric inversion provides complementary application-oriented validation.

Spatial structural consistency was evaluated by comparing each fused band with the selected UAV Red reference component, focusing on high-frequency texture, gradient variation, and edge overlap (Table III). The mean high-frequency correlation, gradient correlation, and Edge $F1$ of the primary spatial validation bands were 0.7975, 0.6128, and 0.6071, respectively. These results demonstrate that the fused products effectively represent fine textures, spatial transitions, and edge structures at the UAV scale.

A spatial gain assessment was further introduced to determine whether DSSAF provides structural information beyond simple interpolation. Directly upsampled PlanetScope imagery was used as the baseline, and its similarity to the UAV spatial reference was compared with that of the fused result (Table IV). The baseline showed near-zero high-frequency and gradient correlations with the UAV reference, indicating that conventional interpolation mainly changes image size without adding real fine-scale spatial information. In contrast, DSSAF introduced UAV-scale structural content while retaining PlanetScope spectral constraints. information with meaningful structural content while maintaining the multispectral constraints of PlanetScope, rather than merely enlarging the image.

B. Bathymetry Accuracy and Geomorphic Patterns

Using the DSSAF-fused high-resolution multispectral imagery and RTK field measurements, a random forest model was constructed to invert shallow-water sandbar topography in terms of elevation and water depth. The effective shallow-water domain was constrained using the water-sandbar classification result, so model training and mapping were limited to optically valid shallow-water conditions [Fig. 6(a)]. The resulting bathymetric reconstruction shows continuous, banded geomorphic structures that correspond closely to the spatial organization of sandbar ridges [Fig. 6(b)], indicating that the fused imagery provides sufficient spatial detail and spectral discrimination for shallow-water topographic retrieval.

Profile-based validation along the cross-shore direction shows that the inverted bathymetric sequences reproduce both the phase and amplitude of elevation variations observed in RTK measurements. Across all validation samples, the random forest model achieved $R^2 = 0.924$, RMSE = 0.079 m,

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TABLE V

EFFECTS OF LAND-COVER STRATIFICATION ON SPECTRAL FIDELITY AND SPATIAL STRUCTURE PRESERVATION IN THE DSSAF FRAMEWORK

Model configuration	Spectral Index		Spatial Index		
	R^2	RMSE	High-frequency Correlation	Gradient Correlation	Edge F1
Non-stratified global regression	0.791	0.138	0.6823	0.5822	0.5907
Land-cover-stratified regression	0.928	0.081	0.8604	0.7128	0.7071

TABLE VI

PERFORMANCE COMPARISON OF DIFFERENT REGRESSION MODELS IN BAND-SPECIFIC SPATIAL PROXY CONSTRUCTION, EVALUATED USING SPECTRAL AND SPATIAL METRICS

Regression model	Spectral Index		Spatial Index		
	R^2	RMSE	High-frequency Correlation	Gradient Correlation	Edge F1
Linear regression	0.705	0.164	0.79	0.6029	0.6087
Support vector regression	0.8	0.135	0.7773	0.5841	0.5902
Gradient boosting regression	0.763	0.147	0.7901	0.5902	0.5973
Random forest regression	0.928	0.081	0.8604	0.7128	0.7071

TABLE VII

ACCURACY COMPARISON OF SHALLOW-WATER BATHYMETRIC INVERSION USING PLANETSCOPE IMAGERY, UAV RGB IMAGERY, AND DSSAF-FUSED IMAGERY

Input imagery	Spatial resolution	Input bands	Train:Test	R^2	RMSE (m)	MAE (m)	Bias (m)	Pearson's r
PlanetScope	3 m	8	7:3	0.582	0.189	0.123	0.042	0.827
UAV RGB	5 cm	3	7:3	0.842	0.117	0.063	0.023	0.933
DSSAF-fused	5 cm	8	7:3	0.924	0.079	0.041	0.013	0.963

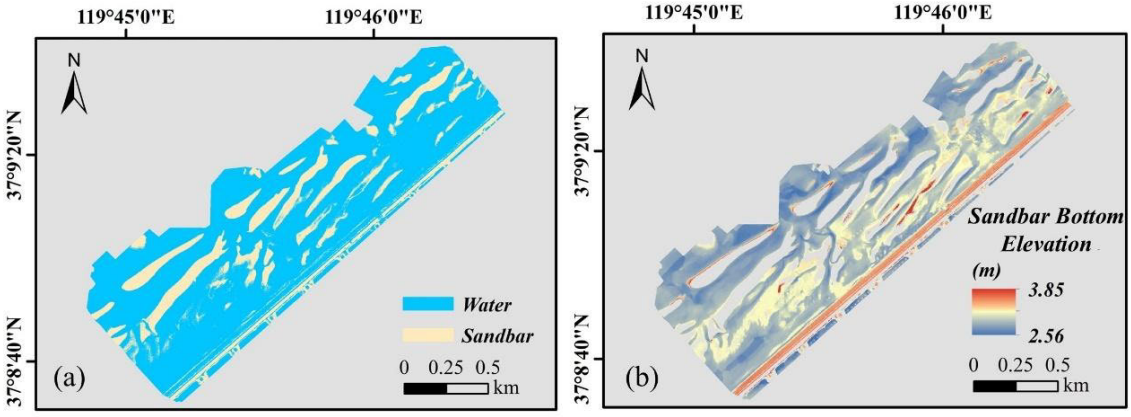


Fig. 6. Spatial results of water depth inversion. (a) Water body–sandbar zoning map. (b) Spatial distribution of the inverted subaqueous sandbar bottom elevation in the shallow-water zone of the sandbar (unit: m).

and MAE = 0.041 m. As illustrated by the profile comparisons (Fig. 7), the inverted curves closely align with the RTK-derived profiles along major relief segments and capture both the overall shoreward elevation increase and multiple

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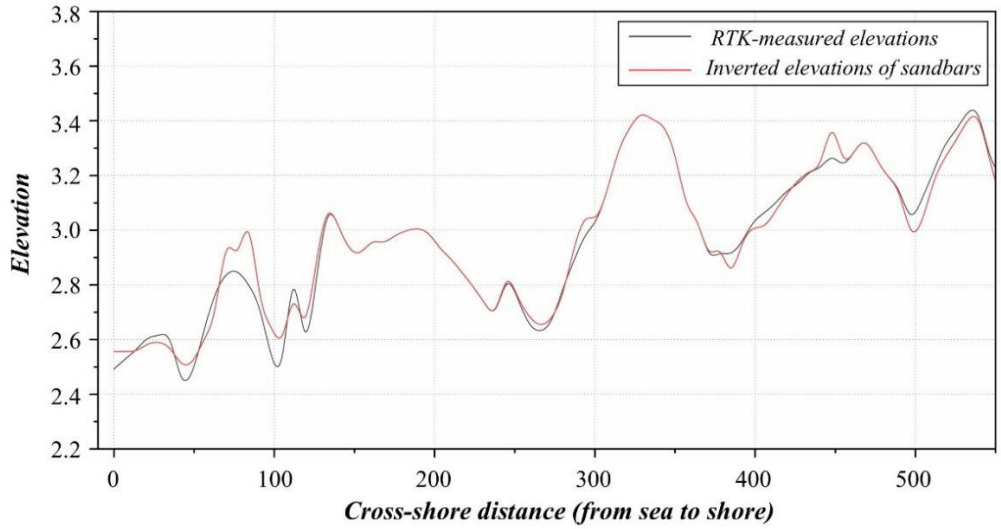


Fig. 7. Validation: Comparison of inverted sandbar elevations with RTK-measured elevations along cross-shore profiles.

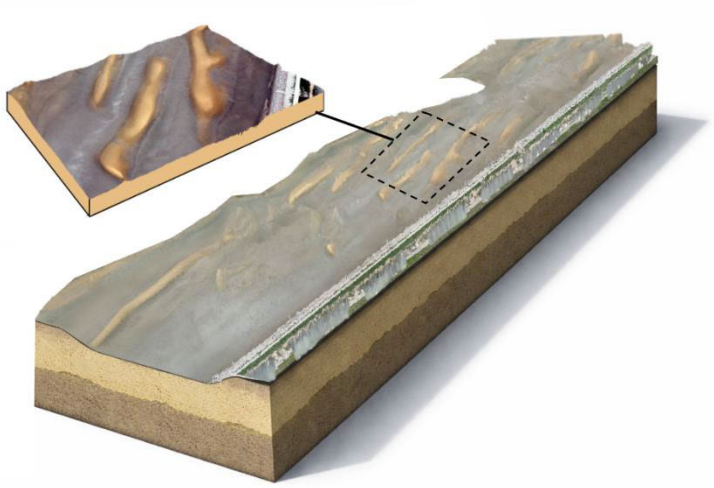


Fig. 8. Reconstructed 3-D sandbar topography accurately captures terrain undulations and is highly consistent with observed field conditions.

secondary undulations. Direct comparison with other satellite-based bathymetry studies should be made cautiously because water depth, substrate type, optical water properties, and reference-data density differ among sites. Nevertheless, the error levels obtained here are substantially lower than those of many machine-learning bathymetry inversions based only on satellite observations. Remaining deviations are mainly concentrated near the lower sandbar, where elevations tend to be slightly underestimated. These errors are consistent with known limitations of optical shallow-water inversion in deeper shallow-water zones with reduced bottom visibility and limited training samples.

Overall, the random forest-based bathymetric inversion using DSSAF-fused imagery achieved high accuracy and strong spatial coherence across the optically shallow-water domain. These results indicate that the proposed fusion approach can support high-resolution water-depth mapping in nearshore sandbar environments and can provide a useful basis for 3-D terrain reconstruction and erosion-deposition assessment (Fig. 8).

V. Discussion

A. Sources of Performance Gain in DSSAF

The land-cover stratification ablation results indicate that spatial–spectral fusion over heterogeneous surfaces cannot rely solely on a single global model (Table V). Compared with global modeling, stratified modeling substantially reduces spectral errors while improving spatial structural metrics, suggesting that spectral–spatial mapping relationships differ markedly among land-cover types. Without land-cover differentiation, a global model must jointly fit water, bare surfaces, wet surfaces, vegetation, and other targets, making it susceptible to mixed pixels and class boundaries. By partitioning the study area into more homogeneous subregions, the stratification strategy stabilizes cross-sensor relationships and reduces pseudo-texture artifacts. Thus, land-cover stratification functions as an important constraint on heterogeneity rather than as a simple preprocessing step.

The comparison among regression models further shows that band-specific spatial proxy construction depends on model

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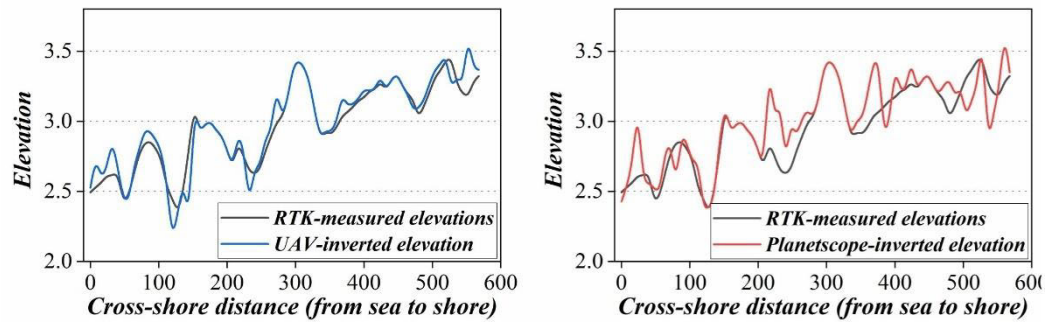


Fig. 9. Comparison of shallow-water bathymetric inversion results derived from different input imagery sources.

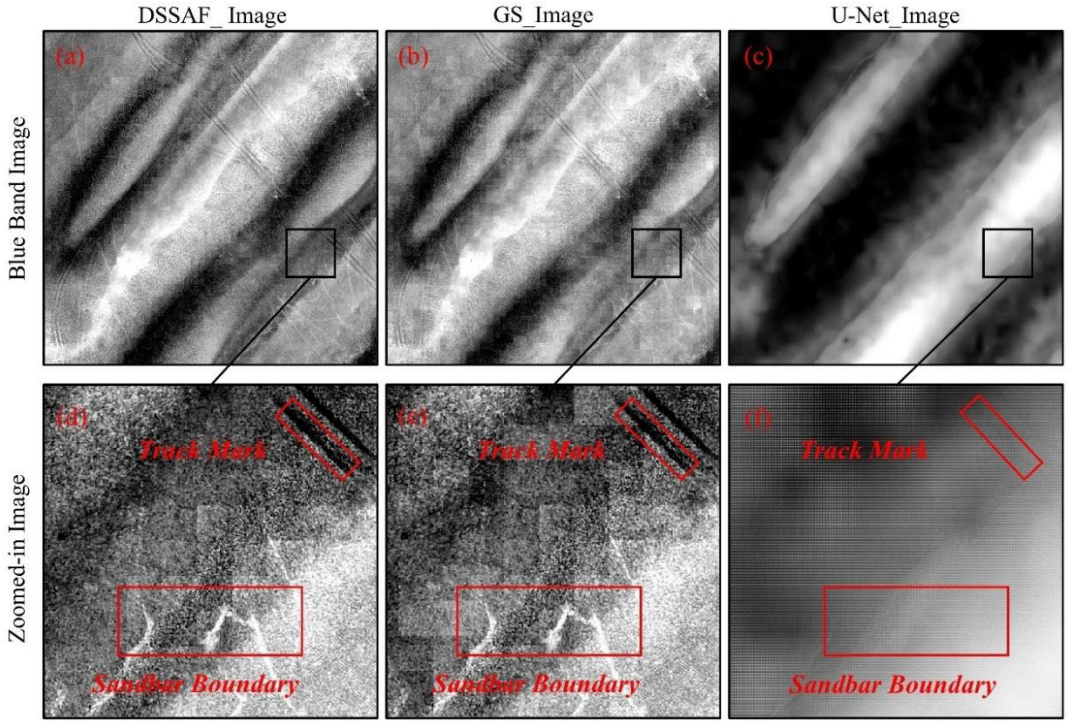


Fig. 10. Visual comparison of reconstructed Blue-band images over the intertidal sandbar area generated by different fusion methods.

representation capacity (Table VI). Random forest regression outperformed linear regression, support vector regression, and gradient boosting regression in both spectral fitting and spatial structure recovery, achieving the highest R^2, the lowest RMSE, and the best high-frequency, gradient, and edge-consistency metrics. This confirms that the relationship between high-resolution spatial features and multispectral target bands is nonlinear and influenced by local texture, land-cover type, radiometric differences, and scale effects. The ensemble-tree structure of random forest regression provides robustness to high-dimensional feature interactions, local heterogeneity, and anomalous samples, making it well suited to band-specific cross-scale modeling.

The downstream bathymetry results demonstrate the practical value of the fused imagery (Table VII). As summarized in Table VII, all three experiments used the same RTK reference samples, the same training–testing partition, and the same Random Forest configuration, whereas spatial resolution and spectral dimensionality differed according to the intrinsic characteristics of each image product. Therefore, the comparison was designed to evaluate the practical spatial–spectral information provided by the three image products under otherwise consistent inversion conditions. PlanetScope-only inversion is constrained by mixed pixels along shallow sandbar margins, land-water transition zones, and areas with rapid local depth variation (Fig. 9). UAV RGB-only inversion performs better because centimeter-scale spatial structure is important for representing shallow-water topography, but its limited spectral dimensionality restricts the characterization of water-column attenuation, substrate differences, and depth-related spectral variations. DSSAF-fused imagery achieved the highest inversion accuracy, with $R^2 = 0.924$ and RMSE = 0.079 m. Relative to PlanetScope-only and UAV RGB-only inputs, RMSE decreased by approximately 57% and 31%, respectively. Under these controlled inversion conditions, the results indicate that the combined fine spatial resolution and multispectral information provided by the DSSAF-fused product offers greater practical information for shallow-water

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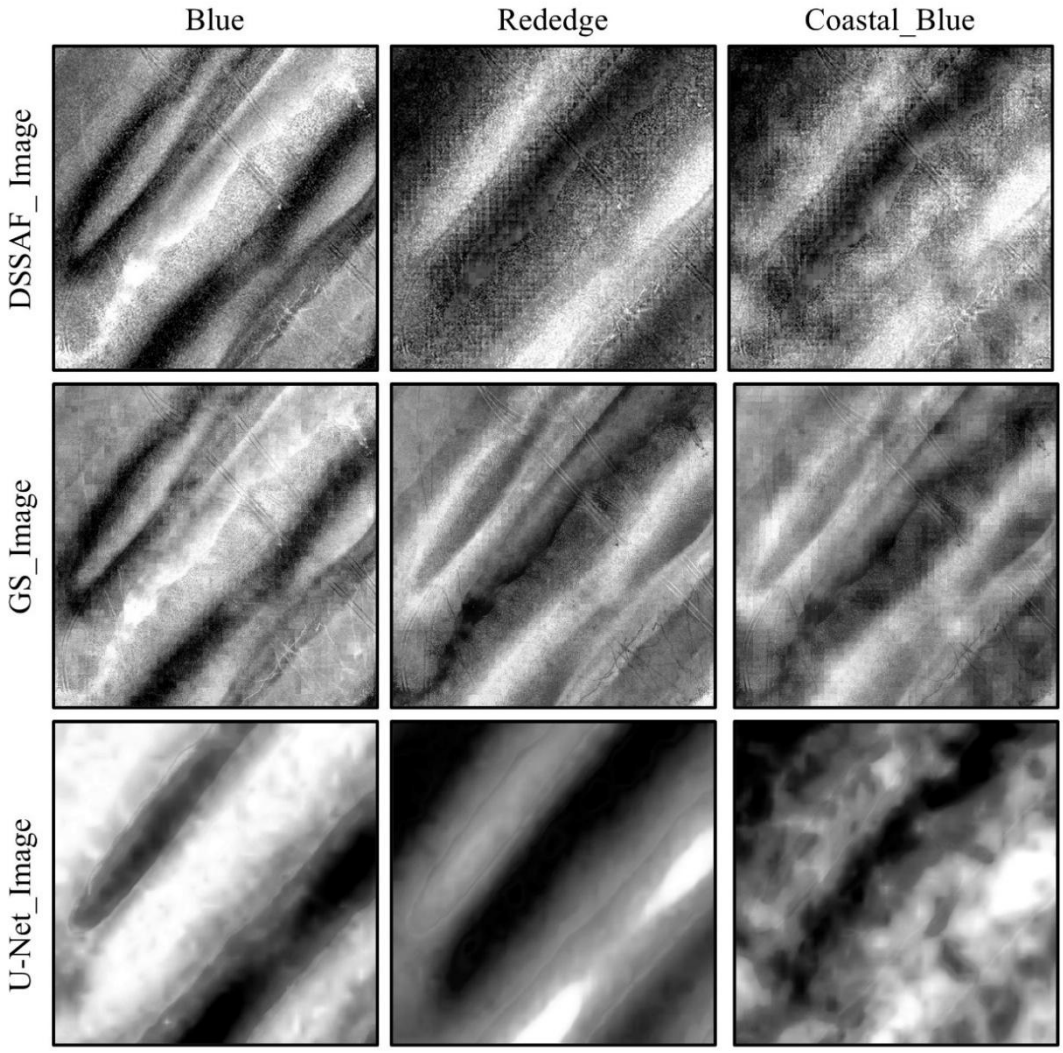


Fig. 11. Interband visual comparison of fused products across representative spectral bands.

bathymetric inversion than either native PlanetScope or UAV RGB imagery alone.

Taken together, these comparative results indicate that the performance gain of DSSAF arises from the joint effects of nonlinear regression, land-cover stratification, and multisource data complementarity. Random forest regression provides stable modeling of complex spatial–spectral relationships, land-cover stratification reduces class-dependent bias over heterogeneous surfaces, and the fused imagery integrates UAV-scale spatial structure with PlanetScope multispectral information. More importantly, these image-level improvements translate into more accurate shallow-water bathymetric inversion, demonstrating the value of DSSAF for quantitative remote-sensing applications rather than visual enhancement alone. Importantly, these image-level improvements can be translated into higher accuracy in downstream water-depth inversion, demonstrating that the proposed method not only improves the visual and structural quality of the fused imagery, but also enhances its applicability for quantitative shallow-water terrain inversion. Therefore, the core value of the DSSAF framework lies not merely in generating multispectral imagery with higher spatial resolution, but in constructing a high-resolution multispectral feature representation that simultaneously incorporates spectral constraints, spatial structural information, and downstream application effectiveness.

B. Mechanism of Method Effectiveness: Balanced Cross-Sensor Conditioning via Stratified Regression and Residual Injection

To further examine the proposed DSSAF framework, the method was compared with two representative fusion baselines: Gram-Schmidt (GS) pan-sharpening and a U-Net-based deep learning model. As shown in Fig. 10, the GS-derived Blue-band image contains noticeable high-frequency artifacts, whereas the U-Net result is overly smoothed and loses microtextures and subtle contrast patterns. DSSAF yields the most visually and radiometrically plausible Blue-band reconstruction among the three methods. This comparison indicates that DSSAF can preserve fine spatial structures while suppressing both spurious textures and over-smoothing.

The interband behavior further clarifies the underlying mechanism. As shown in Fig. 11, GS produces Blue,

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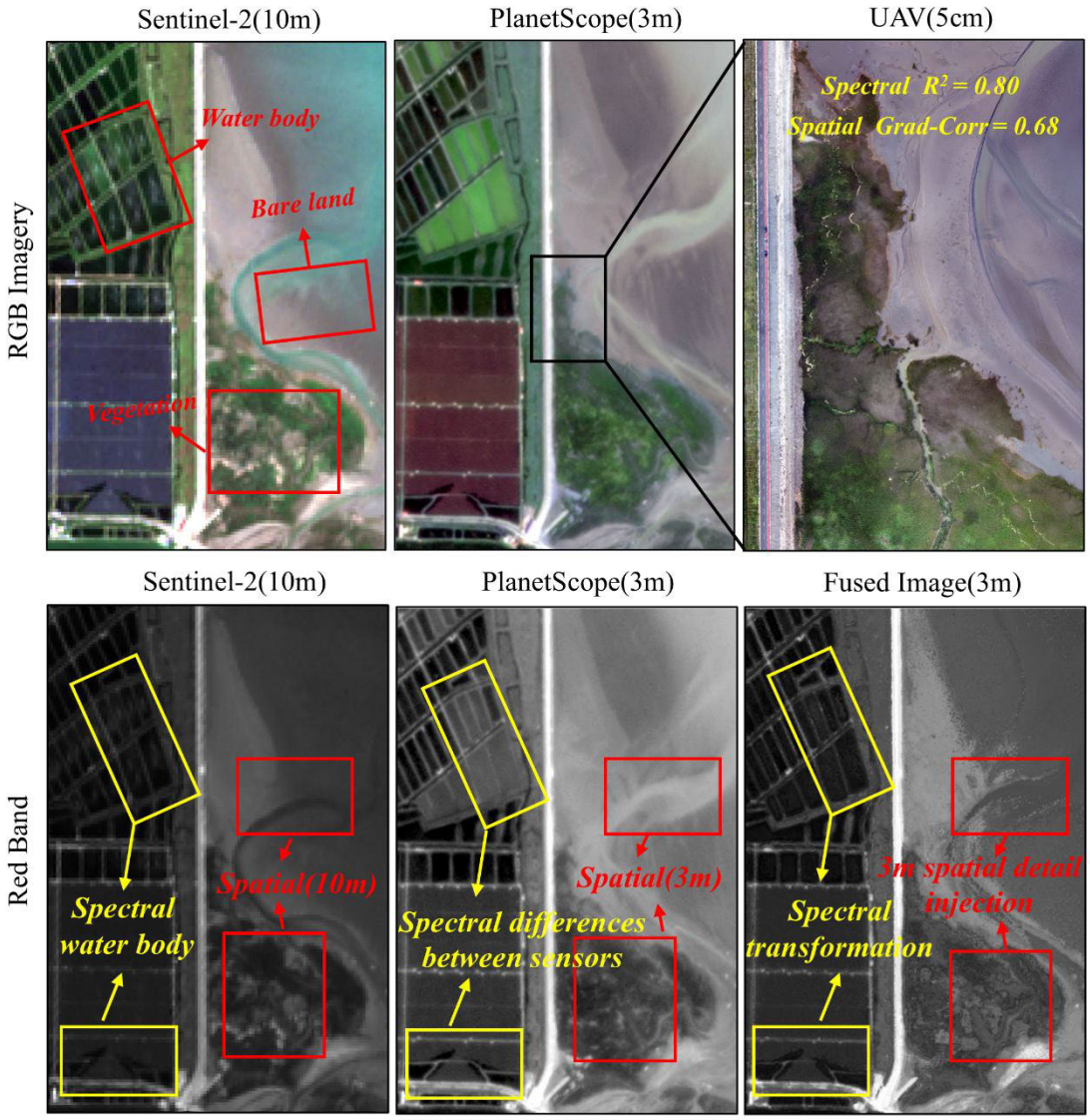


Fig. 12. Cross-sensor transfer validation of the DSSAF framework using PlanetScope–Sentinel-2 fusion and local UAV observations.

Red-edge, and Coastal Blue bands that are excessively similar, indicating weak spectral discrimination and a tendency toward spectral collapse [37]. This behavior is expected because GS pan-sharpening is strongly dominated by the high-resolution panchromatic-like input, whereas the low-resolution PlanetScope multispectral observations act mainly as chromatic adjustment. Consequently, the fused products inherit the spatial pattern of the UAV proxy but provide limited band-specific spectral variability [38]. In contrast, the U-Net model produces more visible interband differences but still shows pronounced smoothing and diminished high-frequency detail, consistent with the dominance of coarse PlanetScope multispectral inputs and interpolation-induced blurring [39].

DSSAF mitigates both limitations by balancing the contributions of the two sensors. The stratified band-wise regression stage learns class-consistent nonlinear mappings that preserve band-specific spectral responses, reducing GS-type over-reliance on a panchromatic proxy and avoiding interband homogenization. The residual injection stage then recovers UAV-grid spatial textures in a controlled manner by injecting locally consistent high-frequency structures while constraining large-scale radiometric drift, thereby alleviating U-Net-type over-smoothing.

C. Proof-of-Concept Cross-Sensor Transfer of the DSSAF Framework

To examine the feasibility of applying DSSAF to a different sensor combination, a proof-of-concept cross-sensor experiment was conducted by extending the framework from the UAV–PlanetScope configuration to a PlanetScope–Sentinel-2 configuration. In this setting, PlanetScope imagery provides spatial structural information at the 3 m scale and fine-scale visible-to-NIR variations, whereas Sentinel-2 provides the target multispectral system and low-frequency spectral constraints. The purpose of this experiment is not to establish comprehensive generalization across sensors, regions, or acquisition conditions, but to examine whether the same spectral–structural fusion principle can be implemented with another pair of complementary sensors.

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Because conventional fusion evaluation often lacks dense high-resolution spectral reference data, local UAV observations were used as an independent validation source for observable bands. As shown in Fig. 12, the Sentinel-2 Red band provides the target-sensor spectral benchmark, but its coarser spatial resolution limits the delineation of road boundaries, vegetation patches, and fine tidal-flat textures. Compared with Sentinel-2, the PlanetScope Red band provides clearer 3 m spatial detail, but radiometric differences between PlanetScope and Sentinel-2 mean that direct substitution may cause spectral shifts. In contrast, the DSSAF-fused result incorporates spatial structure from PlanetScope while preserving the spectral-response characteristics of Sentinel-2, leading to clearer representation of road edges, vegetation-bare-flat transitions, and tidal-flat textures.

The comparison with local UAV observations further confirms that the transfer application of DSSAF does not simply replace Sentinel-2 with high-resolution PlanetScope imagery. Through band-specific mapping, structural detail injection, and scale-consistency correction, DSSAF establishes a coherent representation between the spectral constraints of the target sensor and the spatial structures of the higher-resolution imagery. These results demonstrate the potential of DSSAF to extend from a specific sensor pair to broader multisource remote-sensing fusion scenarios.

The transferability of DSSAF derives from its reliance on structural correspondence rather than sensor-specific radiometric relationships. Although different sensor combinations vary in spectral configuration, spatial resolution, and radiometric characteristics, stable spatial structures often remain observable across scales. By treating structural information and spectral constraints as complementary components, DSSAF reduces dependence on a specific sensor pair and provides a general framework for cross-scale multispectral reconstruction.

VI. Conclusion

This study proposed DSSAF, a scale-consistent framework for high-resolution multispectral reconstruction from heterogeneous remote-sensing observations. By integrating fine-resolution structural information from UAV imagery with multispectral satellite constraints, DSSAF addresses spatial–spectral mismatch, cross-sensor inconsistency, and spectral drift during cross-scale reconstruction. Using an intertidal sandbar system in Yantai, China, as a representative validation site, the framework was evaluated through image reconstruction, structural recovery, ablation experiments, transfer testing, and shallow-water bathymetric inversion. The main conclusions are as follows.

1) DSSAF enhances high-spatial-resolution structural information while maintaining multispectral consistency. After reaggregation to the native PlanetScope scale, the fused products remained highly consistent with the original multispectral observations, confirming the effectiveness of the scale-consistency constraint in limiting low-frequency spectral drift. The fused imagery also improved high-frequency texture, gradient structure, and edge consistency, demonstrating that DSSAF introduces meaningful UAV-scale spatial information rather than merely increasing display resolution.
2) Ablation experiments show that the performance gain arises from the interaction among land-cover-conditioned modeling, nonlinear spectral mapping, and structural detail reconstruction. Land-cover stratification mitigates mixed-pixel and class-boundary effects among water, wet surfaces, bare surfaces, vegetation, and other land-cover types. Comparisons among regression models further show that random forest regression provides the most stable spectral fitting and spatial structure recovery, indicating that the mapping between high-resolution features and multispectral target bands is highly nonlinear and surface-dependent.
3) Shallow-water bathymetric inversion confirms the practical value of DSSAF-fused imagery. Compared with PlanetScope-only imagery, DSSAF alleviates mixed-pixel effects caused by coarse satellite resolution. Compared with UAV RGB-only imagery, it supplements richer multispectral information for representing water-column optical variability, substrate differences, and local depth gradients. The DSSAF-based inversion achieved the highest accuracy among the tested inputs, demonstrating that image-level spatial–spectral enhancement can translate into improved quantitative inversion performance.
4) The PlanetScope–Sentinel-2 experiment provides proof-of-concept evidence for cross-sensor implementation of DSSAF. When PlanetScope provides higher-resolution spatial structure, and Sentinel-2 provides the target multispectral system and low-frequency spectral constraints, the same spectral–structural fusion principle remains applicable under the tested conditions. This result demonstrates the feasibility of extending DSSAF beyond the UAV–PlanetScope configuration, although broader generalization across sensor systems, regions, seasons, and land-cover conditions requires further validation.

Overall, DSSAF provides an interpretable and transferable framework for integrating structural and spectral information across observation scales. Its significance lies not only in generating high-resolution multispectral imagery but also in establishing a scale-consistent representation that links image reconstruction with downstream quantitative applications. These findings suggest a potential pathway for cross-sensor and cross-scale environmental mapping, while further multiregion and multitemporal evaluations are required to establish its broader applicability.

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References

[1] J. He, Q. Yuan, J. Li, Y. Xiao, and L. Zhang, "A self-supervised remote sensing image fusion framework with dual-stage self-learning and spectral super-resolution injection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 131–144, Oct. 2023.

[2] K. Zhang et al., "Panchromatic and multispectral image fusion for remote sensing and Earth observation: Concepts, taxonomy, literature review, evaluation methodologies and challenges ahead," *Inf. Fusion*, vol. 93, pp. 227–242, May 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:28:20 UTC from IEEE Xplore. Restrictions apply.

[3] M. Cheng, W. He, Z. Li, G. Yang, and H. Zhang, "Harmony in diversity: Content cleansing change detection framework for very-high-resolution remote-sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 1–19, Dec. 2024.

[4] D. Wen et al., "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021.

[5] L. Wan, Y. Ryu, B. Dechant, J. Lee, Z. Zhong, and H. Feng, "Improving retrieval of leaf chlorophyll content from Sentinel-2 and Landsat-7/8 imagery by correcting for canopy structural effects," *Remote Sens. Environ.*, vol. 304, Apr. 2024, Art. no. 114048.

[6] H.-F. Yan, Y.-Q. Zhao, J. C.-W. Chan, S. G. Kong, N. Ei-Bendary, and M. Reda, "Hyperspectral and multispectral image fusion: When model-driven meet data-driven strategies," *Inf. Fusion*, vol. 116, Apr. 2025, Art. no. 102803.

[7] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[8] J. Ju et al., "The harmonized Landsat and Sentinel-2 version 2.0 surface reflectance dataset," *Remote Sens. Environ.*, vol. 324, Jul. 2025, Art. no. 114723.

[9] K. Obata and H. Yoshioka, "Unmixing-based radiometric and spectral harmonization for consistency of multi-sensor reflectance time-series data," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 396–411, Jun. 2024.

[10] S. Xie, L. Sun, L. Liu, and X. Liu, "Global cross-sensor transformation functions for Landsat-8 and Sentinel-2 top of atmosphere and surface reflectance products within Google Earth Engine," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409609.

[11] L. Xu et al., "Time series analysis for global land cover change monitoring: A comparison across sensors," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112905.

[12] M. Claverie et al., "The harmonized Landsat and Sentinel-2 surface reflectance data set," *Remote Sens. Environ.*, vol. 219, pp. 145–161, Dec. 2018.

[13] D. He, Q. Shi, J. Xue, P. M. Atkinson, and X. Liu, "Very fine spatial resolution urban land cover mapping using an explicable sub-pixel mapping network based on learnable spatial correlation," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113884.

[14] C. Zhang, Q. Wang, and P. M. Atkinson, "Unsupervised object-based spectral unmixing for subpixel mapping," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114514.

[15] Y. Yuan, Z. Lin, X. Jiang, and Z. Fan, "Hyper-parameter optimization-based multi-source fusion for remote sensing inversion of non-photosensitive water quality parameters," *Int. J. Remote Sens.*, vol. 45, no. 18, pp. 6838–6859, Sep. 2024.

[16] Z. Chen et al., "Tradeoffs among multi-source remote sensing images, spatial resolution, and accuracy for the classification of wetland plant species and surface objects based on the MRS_DeepLabV3+ model," *Ecological Informat.*, vol. 81, Jul. 2024, Art. no. 102594.

[17] W. Tang et al., "UAV-borne RGB image and LiDAR fusion for reconstruction of 3D simulated hyperspectral data for crop growth parameter estimation," *Comput. Electron. Agricult.*, vol. 239, Dec. 2025, Art. no. 111058.

[18] F. Nex et al., "UAV in the advent of the twenties: Where we stand and what is next," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 215–242, Feb. 2022.

[19] H. T. Kieu, H. Y. Pak, H. L. Trinh, D. S. C. Pang, E. Khoo, and A. W.-K. Law, "UAV-based remote sensing of turbidity in coastal environment for regulatory monitoring and assessment," *Mar. Pollut. Bull.*, vol. 196, Nov. 2023, Art. no. 115482.

[20] J. He, S. Zhang, X. Cui, and W. Feng, "Remote sensing for shallow bathymetry: A systematic review," *Earth-Sci. Rev.*, vol. 258, Nov. 2024, Art. no. 104957.

[21] J. Jiang et al., "Combining UAV and Sentinel-2 satellite multi-spectral images to diagnose crop growth and n status in winter wheat at the county scale," *Field Crops Res.*, vol. 294, Apr. 2023, Art. no. 108860.

[22] N. Liu, L. Li, W. Li, R. Tao, J. E. Fowler, and J. Chanussot, "Hyperspectral restoration and fusion with multispectral imagery via low-rank tensor-approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7817–7830, Sep. 2021.

[23] Z. Yin, X. Li, P. Wu, J. Lu, and F. Ling, "CSSF: Collaborative spatial–spectral fusion for generating fine-resolution land cover maps from coarse-resolution multi-spectral remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 226, pp. 33–53, Aug. 2025.

[24] M. Ciotola, S. Vitale, A. Mazza, G. Poggi, and G. Scarpa, "Pansharpening by convolutional neural networks in the full resolution framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408717.

[25] J.-L. Xiao, T.-Z. Huang, L.-J. Deng, Z.-C. Wu, X. Wu, and G. Vivone, "Variational pansharpening based on coefficient estimation with nonlocal regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406115.

[26] B. Pan, Q. Qu, X. Xu, and Z. Shi, "Structure–color preserving network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520512.

[27] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.

[28] G. Vivone et al., "Deep learning in remote sensing image fusion: Methods, protocols, data, and future perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 1, pp. 269–310, Mar. 2025.

[29] F. Trevisiol, E. Mandanici, A. Pagliarani, and G. Bitelli, "Evaluation of Landsat-9 interoperability with Sentinel-2 and Landsat-8 over Europe and local comparison with field surveys," *ISPRS J. Photogramm. Remote Sens.*, vol. 210, pp. 55–68, Apr. 2024.

[30] E. Pastucha, E. Puniach, W. Gruszczyński, P. Ćwia
kała, W. Matwij, and H. S. Midtiby, "Relative radiometric normalisation of unmanned aerial vehicle photogrammetry-based RGB orthomosaics," *Photogramm. Rec.*, vol. 37, no. 178, pp. 228–247, Jun. 2022.

[31] R. L. S. Dias, R. S. S. Amorim, D. D. da Silva, E. I. Fernandes-Filho, G. V. Veloso, and R. H. F. Macedo, "Relative radiometric normalization for the PlanetScope nanosatellite constellation based on Sentinel-2 images," *Remote Sens.*, vol. 16, no. 21, p. 4047, Oct. 2024.

[32] M. T. Eismann and R. C. Hardie, "Hyperspectral resolution enhancement using high-resolution multispectral imagery with arbitrary response functions," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 455–465, Mar. 2005.

[33] X. Zhang et al., "Evaluation of BRDF information retrieved from time-series multiangle data of the Himawari-8 AHI," *Remote Sens.*, vol. 14, no. 1, p. 139, Dec. 2021.

[34] Y. Zhao and D. Liu, "A robust and adaptive spatial–spectral fusion model for PlanetScope and Sentinel-2 imagery," *GIScience Remote Sens.*, vol. 59, no. 1, pp. 520–546, Dec. 2022.

[35] T. Miura, A. Huete, and H. Yoshioka, "An empirical investigation of cross-sensor relationships of NDVI and red/near-infrared reflectance using EO-1 hyperion data," *Remote Sens. Environ.*, vol. 100, no. 2, pp. 223–236, Jan. 2006.

[36] M. J. Mabula, D. Kisanga, and S. Pamba, "Application of machine learning algorithms and Sentinel-2 satellite for improved bathymetry retrieval in lake Victoria, Tanzania," *Egyptian J. Remote Sens. Space Sci.*, vol. 26, no. 3, pp. 619–627, Dec. 2023.

[37] J. Duran, A. Buades, B. Coll, C. Sbert, and G. Blanchet, "A survey of pansharpening methods with a new band-decoupled variational model," *ISPRS J. Photogramm. Remote Sens.*, vol. 125, pp. 78–105, Mar. 2017.

[38] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[39] M. Sdraka et al., "Deep learning for downscaling remote sensing images: Fusion and super-resolution," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 202–255, Sep. 2022.

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